Robust Learning Under Label Noise for Real-World Classification

Evaluation & Reproducibility Report | Experiment ID: 37c309b9-cf03-42d5-9dbd-e63d9a98c74b | Generated: 2026-10-07 00:26:22

1. Dataset & Experimental Setup

Target Variable:	diagnosis	**Classes:**	2 (Benign, Malignant)
Total Rows:	569	**Preprocessed Features:**	30
Train / Test Split:	397 (70%) / 86 (15%)	**Noise Levels Evaluated:**	0%, 20%
Random Seed:	[42]	**Noise Strategy:**	Symmetric

2. Model Evaluation Across Noise Levels

Evaluated strictly on the clean holdout test set to measure genuine generalization ability under synthetic training noise.

Model	Type	Noise	Accuracy	Precision	Recall	F1-Score	Time (s)
Baseline Logistic Regression	Baseline	0%	0.9884	0.9886	0.9884	**0.9883**	0.02
Robust Sample-Weighted (Logistic Regression)	Robust	0%	0.9884	0.9886	0.9884	**0.9883**	0.01
Baseline Logistic Regression	Baseline	20%	0.9651	0.9656	0.9651	**0.9652**	0.02
Robust Sample-Weighted (Logistic Regression)	Robust	20%	0.9767	0.9781	0.9767	**0.9769**	0.02

3. Confident Learning Label Issue Detection

Cross-validation out-of-fold probability estimation identified **59** suspicious training samples (14.86% of training split) exceeding the issue threshold of 0.50. These samples exhibited substantial margin disagreement between the assigned label and the learned distribution.

Row ID	Observed Label	Predicted Class	Confidence	Observed Prob	Issue Score	Status
376	Malignant	Benign	0.97	0.03	**0.95**	Potential Issue
323	Benign	Malignant	0.96	0.04	**0.95**	Potential Issue
219	Benign	Malignant	0.93	0.07	**0.90**	Potential Issue
252	Benign	Malignant	0.93	0.07	**0.90**	Potential Issue
391	Malignant	Benign	0.93	0.07	**0.89**	Potential Issue
540	Malignant	Benign	0.92	0.08	**0.88**	Potential Issue
339	Benign	Malignant	0.92	0.08	**0.88**	Potential Issue
565	Benign	Malignant	0.92	0.08	**0.88**	Potential Issue

4. Academic Findings & Discussion

• **Baseline Performance:** Under pristine labels (0% noise), Baseline Logistic Regression achieved the highest F1-score of 0.9883 (Accuracy: 0.9884).

• **Noise Robustness:** At 20% label noise, the robust model (Robust Sample-Weighted (Logistic Regression)) demonstrated superior noise resilience, achieving an F1-score of 0.9769 compared to Baseline Logistic Regression's 0.9652 (+0.0117 improvement).

5. Methodological Notes & Reproducibility

Reproducibility: Results are strictly deterministic given the configured random seed and split ratios. Label noise was applied solely to the training set; test labels remained unaltered to ensure valid evaluation.
Academic Note: Confident learning scores indicate statistical discrepancy between observed labels and feature density; they represent potential label noise, not mathematical proof of label error in uncontrolled environments.